

ICA2_Group106

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A. Dataset and Objective

In 2016, the EU referendum took place in the UK to decide whether or not to leave the EU. Data from 1070 wards were shared, and we are going to explore how the proportion of “Leave” votes varies depending on the Ward. We will study the dataset “ReferendumResults.csv”, which contains entries for 1070 wards. There is a total of 49 variables, including the number of Total votes and Leave votes of the ward.

The first 803 rows (wards) have the number of Leave votes recorded, while the remaining 267 rows have **Leave** set to -1 (treated as missing, we converted to NA). We created **LeaveProp**, the response variable, corresponding to the proportion of Leave votes in each ward.

The dataset provides a rich set of demographic, socioeconomic, procedural and geographic covariates for each ward, both categorical and numerical (derived from the 2011 UK Census and other sources). We then explore these covariates to build a data-driven, theoretically justified model.

B. Exploratory Data Analysis

Our first task was to assess the structure of our dataset by investigating linear relationship to the response variable, identify outliers and interrelationships of each variable.

Geographic Looking at geographic variables, we concluded that **RegionName**, **AreaType** are categorical variables which we can implement into model building.

We implemented hierarchical clustering into Regions, using **ClusterGroup** to simplify the regions into similar characteristics in order to assess the relationship between each variables in different groups.

The dendrogram reveals four main clusters among the nine regions, with London separated at a high level of ‘Separation.’ This implies that London’s profile is distinct, while the remaining regions group into three additional clusters, suggesting moderate similarities within each cluster. Their weak direct association led us to explore interactions with socio-demographics.

For **AreaType** vs **LeaveProp**, the boxplot indicates that E09 (London) has the lowest mean Leave vote proportion (0.41), while E08 (metropolitan districts) exhibits the highest mean (0.58). We conclude that these categorical variables hold no linear significance on its own, however we may categories some variables and form interactions.

Finally, the continuous variable **Density** was concluded to not be strictly linear to the response variable.

Demographic We merged age groups into `ChildPct`, `YoungPct`, `AdultPct` and `OlderPct` accordingly. After plotting its residuals against the variable. We concluded that only `OlderPct`, including `MeanAge` and `AdultMeanAge` inhibit a linear relationship to the response variable.

We also noticed that age groups inhibit a linear relationship when interacting with regional variables. With reasonable linearity of `ClusterGroup*AdultPct` with the response, specifically showing for `AdultAge*ClusterGroup` a downtrend in London.

In terms of Ethnicity, the variables alone don't explain much, we approached it by breaking it down into PC1, PC2 and PC3, (Fig 4.1):

The correlation table shows strong anti-correlation between White and Minority groups ($r = -0.69$ to -0.89). However, Minorities exhibit significant differences (e.g., Pakistani and Indian correlation = 0.24). Since ethnicities sum to 100%, fitting them separately would cause multicollinearity. We use PCA, with the first two components explaining 85% of the variance. PC1 contrasts White-majority vs. Minority-majority, and PC2 distinguishes Black vs. Pakistani minorities (-0.81 vs. 0.50). PC3, capturing 15% of the variance, contrasts Indian vs. Pakistani (0.77 vs. -0.58). These components are key for our regression analysis. We will use PCA1, PCA2, PC3, but also interchange with other ethnicities in our model.

Socioeconomic We notice strong linearity with response variable for education: `NoQuals`, `L1Quals` and `L4Quals_plus` with no notable outliers. Similarly for occupation variables, `Residents` and `Households`, we notice no departure from linearity and no major outliers when plotting Residents against the response, apart from `Students` showing a clear departure from linearity.

Decomposing `C1C2DE` and `C2DE` into `C1`, `C2`, and `DE` revealed non-linear residual patterns for all three.

We found no evidence of non-linearity or significant outliers among the housing variables (`Owned`, `OwnedOutright`, `SocialRent`, `PrivateRent`), deprivation variables (`MultiDepriv`, `Deprived`) and occupation variables (`Unemp`, `UnempRate_EA`, `HigherOccup`, `RoutineOccupOrLTU`), indicating that their relationships with the outcome variable appear to be well-behaved and appropriately modeled.

Procedural Postal voting status is our only procedural covariate, shows a slight reduction in Leave share but adds little explanatory power. Although the variable did not show any huge variation in trend, we will consider this variable in the model.

Summary We concluded no huge notable outliers in the model, as we have some variables which are not linear to the model, we may consider building a GLM/GAM where we can utilise some of these variables. Given the nature of the variable `LeaveProp`, we will first hypothesise an implementation of a logistic regression. We tested predictor linearity against the logit using the Box-Tidwell method.

Linearity to logit

The response variable, `LeaveProp`, is bounded between 0 and 1, making a logit model appropriate. Some predictors exhibit non-linear relationships with `LeaveProp`, so it's essential to check whether these variables are linear to the logit function. Based on this, we will decide whether to use a Generalized Linear Model (GLM) with a logit link or a Generalized Additive Model (GAM) to capture the non-linear effects.

For this we used the Box-Tidwell Test to test for departures from linearity.¹

The tests assumes non-linearity to the logit under p-value 0.05 for: `ClusterGroup`, `ChildPct`, `AdultPct`, `OlderPct`, `Indian`, `Students`, `NoQuals`. Therefore we will be proceeding with a GAM to give us the flexibility to use smooth functions on these variables, which GLM cannot capture.

¹Barbara G.Tabachnick, Lina S.Fidell (2012). "Using Multivariate Statistics" (6th Edition)

C. Model Building

Model selection and strategy

We chose GAM with a binomial family (logit link) for modeling Ward-level Leave Proportions. This gives us the flexibility to utilise variables that aren't linear to the logit function, this approach commonly used in similar situations.²

We will use `weights = NVotes` in our model, since `LeaveProp` is a proportion based on a different number of voters across the wards.

We used forward selection on AIC, then tested pairwise interactions. As said in our EDA, there are some variables assumed to be non-linear in respect to the logit of `LeaveProp`, we will use them by implementing the spline function. Finally, when we carefully add variables to the model, we will be wary of any potential collinearity issues.

Phase 1: Variable Addition and Collinearity

We followed a forward-selection loop: insert the most significant covariate, compare AIC, inspect the correlation matrix, removed any variable pair whose VIF exceeded 10, then reassess the candidate pool.

To read the correlation matrix heatmap (Fig 4.2), the deep red cells indicate a perfect positive correlation of +1, deep blue cells a perfect negative correlation of -1, and white cells signify no correlation (0).

To start with our model building, we identified that `L1Quals1` is the variable that yields the lowest p-value and covers the most deviance (64.8%) out of all. We decide to create an initial model `model0` with this single variable, with an adjusted R^2 of 0.655 and AIC of 145351. We further repeat this process.

Adding eight covariates — `s(OlderPct)`, `L4Quals_plus`, `PC3`, `White`, `RegionName`, `Asian`, `Density`, `Residents` — slashed the intercept-only AIC from 155,847 to 52,333.8 and accounted for 88.4% of the deviance (adj- R^2 =0.886).

During this step the correlation matrix (Fig 4.3) highlighted two collinear pairs:

Residents vs Households (VIF 13.1), `RegionName` vs `AreaType` (perfect overlap)

Side-by-side models retained Residents (AIC 48,734 < 49,181) and `RegionName` (47,455 < 56,286); the higher-AIC partners were dropped before the next pass.

The next variables that were statistically significant were the socioeconomic variables. Incorporating `OwnedOutright`, `NoQuals`, `Unemp`, `C2DE`, `MultiDepriv` drove AIC down to 43,286.46 and lifted deviance explained to 90.7% (adj- R^2 =0.908).

While implementing VIF checks, it exposed us to two further pairings above the threshold:

`RoutineOccupOrLTU` vs `C2DE` (VIF 15.7), `MultiDepriv` vs `Deprivd` (VIF 20.2)

Lower-AIC terms — `RoutineOccupOrLTU` (43,074 < 43,286) and `MultiDepriv` (43,286 < 43,512) — were kept, leaving with 15 covariates. Additional single terms only removed <2% off AIC, so Phase 1 concluded here under the parsimony rule.

```
## LeaveProp ~ L1Quals + s(OlderPct) + L4Quals_plus + PC3 + White +  
##               Asian + Density + C2 + Residents + RegionName + OwnedOutright +  
##               NoQuals + Unemp + C2DE + MultiDepriv  
##  
## Deviance explained: 90.7%  
## AIC: 43286.46
```

²Barbara G. Tabachnick, Lina S. Fidell (2012). "Using Multivariate Statistics" (6th Edition)

Phase 2: Interactions

The Phase 2 of this building process will consist on considering possible interactions between the covariates of our model. By adding interactions, the goal is to lower the AIC and increase the accuracy of our model.

From our EDA we noticed that the age group variable `YoungPct` yield linearity when interacting with `AreaType`, potentially yielding lower AIC. However, we previously chose to keep `AreaType` over `RegionName`, due to collinearity issues. We studied the potential power of using the interaction instead of `RegionName`, but we still obtain better results with this last single variable.

We systematically checked potential models by adding possible interactions and comparing their AIC difference in respect to the previous one:

We found that the interaction between `White` and `OwnedOutright` lowered the AIC the most, following the system defined, we obtain:

```
## Previous model AIC:  
## 43286.46
```

```
## New model AIC:  
## 41483.99
```

```
## The difference in AIC with the previous model is:  
## 1802.475
```

The interaction `White:OwnedOutright` lowered AIC by 1,802, showing that Leave support is highest in wards combining a large White population with outright home ownership (a proxy for older, settled voters).

We found that the next interaction that lowered the AIC the most was the one between `PC3` and `L4Quals_plus`, obtaining the following:

```
## Previous model AIC:  
## 41483.99
```

```
## New model AIC:  
## 39944.57
```

```
## The difference in AIC with the previous model is:  
## 1539.419
```

`PC3:L4Quals_plus` cut AIC by 1,539, indicating the pro-Remain effect of higher education strengthens where Indian residents outnumber Pakistanis.

By contrast, `Unemp:MultiDepriv` removed only 114 AIC points—below our 2% threshold—and its signal is already absorbed by the main effects of unemployment and deprivation, so it was excluded on parsimony grounds.

Smooth Term Tuning

We readjust the smooth term for `OlderPct` such that given the concurvity values for the smooth term `OlderPct` (estimate = 0.057, observed = 0.126), the low values indicate minimal redundancy with other predictors, suggesting it contributes uniquely and reliably to the model, while also lowering the AIC as shown below.

```
## Previous model AIC:  
## 39944.57
```

```
## New model AIC:  
## 38049.7
```

```
## The difference in AIC with the previous model is:  
## 1894.87
```

Our final model is shown below:

```
## Family: binomial  
## Link function: logit  
##  
## Formula:  
## LeaveProp ~ L1Quals + s(OlderPct, bs = "cr", k = 50, sp = 0.5) +  
##           PC3 + L4Quals_plus + White + Asian + Density + C2 + Residents +  
##           RegionName + OwnedOutright + NoQuals + Unemp + C2DE + MultiDepriv +  
##           (White:OwnedOutright) + (PC3:L4Quals_plus)  
##  
## Model fit:  
## Adjusted R-squared: 0.917  
## Deviance explained: 92.1%  
## AIC: 38049.7
```

D. Model fit and limitations

Overall, the logistic GAM satisfies its core assumptions for the vast majority of wards, exhibiting only minor deviations in extreme cases and slight under-dispersion.

We checked model fit against four core assumptions of a logistic GAM. First, the deviance-residual QQ-plot and residuals-versus-fitted scatter (Figs 3.1–3.2) confirm that the logit link and smooth functions capture the main trends, with only the most extreme wards (Leave < 25% or > 75%) showing slight tail departures. Second, the residual histogram (Fig 3.3) is centred and symmetric, indicating no systematic bias. Third, the observed-versus-fitted curve (Fig 3.4) shows that most of our values align and are close to the straight line, reflecting on our accuracy. Fourth, concurvity diagnostics verify that each smooth term is identifiable and well-supported by the data. The dispersion estimate of 0.0077, well below the theoretical value of 1, reveals mild under-dispersion, consistent with slight over-smoothing. We also assume independence across wards, although spatial correlation may remain unmodelled.

Remaining limitations include minor non-linear effects in a few covariates, modest concurvity among smooth terms, under-dispersion, and potential spatial dependence. Future refinements could relax smoothing penalties, include additional interactions, or incorporate spatial random effects.

E. Conclusion

The final logistic GAM, incorporating 15 main effects and two interactions, explains 92% of the variation in ward-level Leave support (AIC 38,050). Residual misfit is limited to a small number of extreme wards.

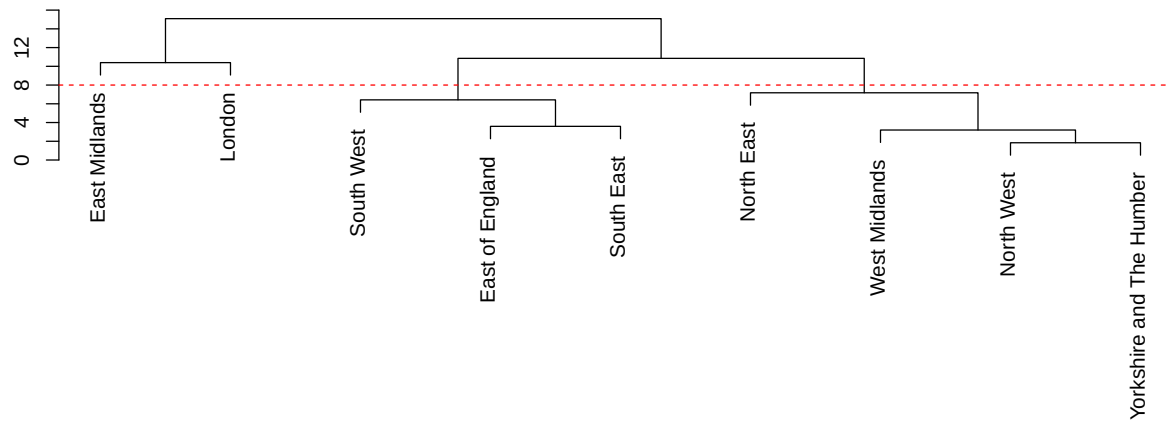
Education is the primary determinant: a greater share of residents with basic or no qualifications is associated with higher Leave support, whereas a higher proportion of degree holders reduces it. This educational

effect is strongest in wards where Indian populations outnumber Pakistani populations, as captured by the positive `PC3:L4Quals_plus` interaction. The negative main effect of `PC3` indicates that wards with a larger Pakistani presence are more Leave-leaning, even before accounting for education levels. Ethnic and tenure dynamics also influence outcomes: although outright home ownership is mildly Remain-leaning overall, in predominantly White wards, the `White:OwnedOutright` interaction drives a stronger Leave preference. Additionally, a modest positive coefficient on the Asian population suggests a slight pro-Leave tendency in mixed-ethnicity areas after controlling for other factors. Higher population density is associated with lower Leave support, reaffirming the stronger Remain tendency in urban areas such as London. Age structure, regional differences, and socioeconomic strain—captured by `C2/DE` social grades, unemployment, and multiple deprivation—further contribute to Leave support, though with smaller marginal effects.

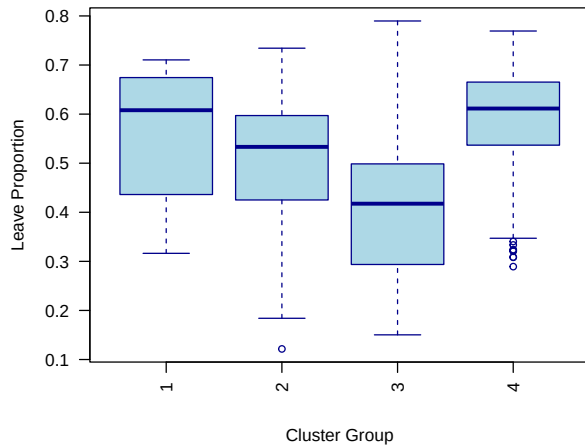
The referendum outcome is explained by intersecting structural axes: education, ethnicity linked with tenure, urban form, age distribution, regional location, and socioeconomic status. Older, less qualified, predominantly White communities outside major urban centers were more likely to support Leave, while more diverse, highly educated urban populations predominantly supported Remain. Model errors are concentrated among spatially adjacent or demographically atypical wards, indicating future improvements could come from incorporating spatial random effects or using updated demographic data.

Graphs

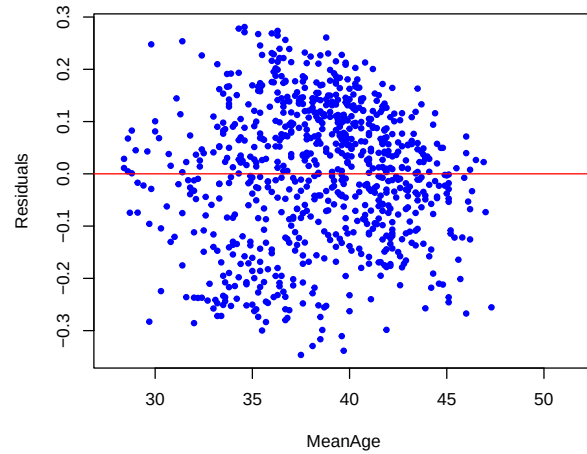
1 Hierarchical Clustering Dendrogram



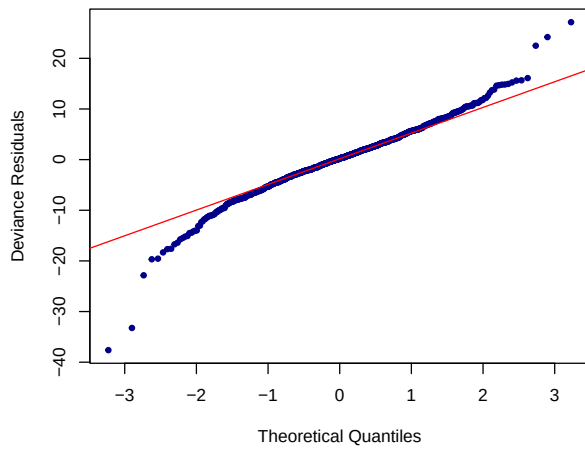
2.1 Leave Proportion by Cluster Group



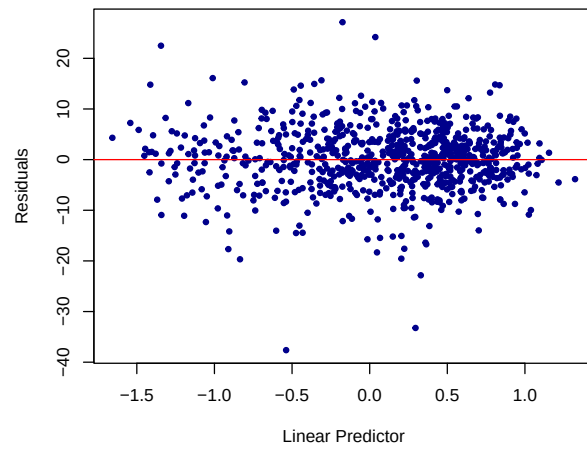
2.2 Residuals vs MeanAge



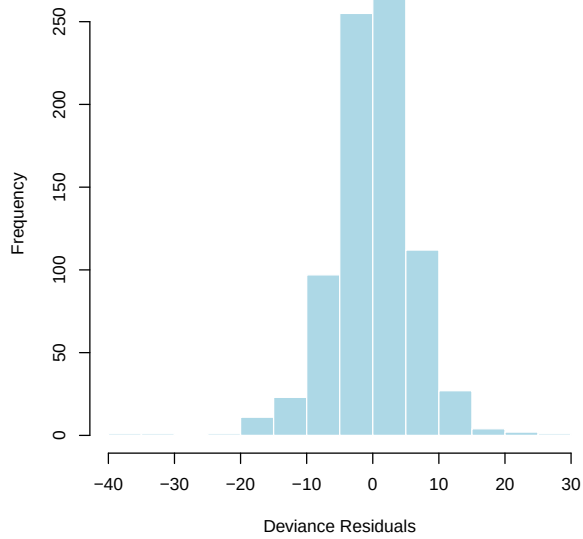
3.1 QQ-Plot of Deviance Residuals



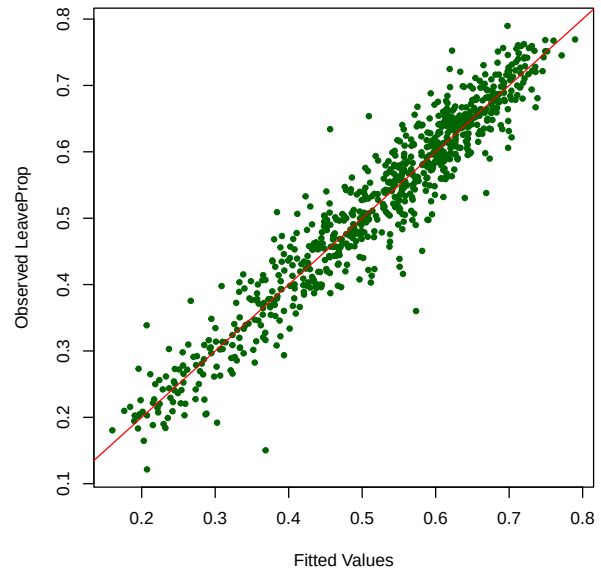
3.2 Residuals vs Linear Predictor



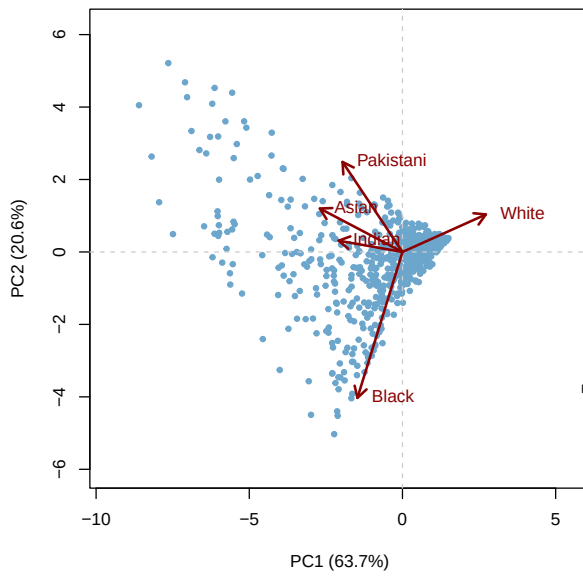
3.3 Histogram of Deviance Residuals



3.4 Observed vs Fitted Values



4.1 PCA Biplot for Ethnicity Variables



4.2 Correlation Heatmap of Predictors

